go-On — Master Timetable Data Integrity & Portal Consistency Audit

End-to-End Architectural Audit, Root Cause Resolution, Network Failover & Device Verification

Target Device: Samsung Galaxy S24 (SM-S921E)	**Database State:** Authoritative SQLite (42 Periods/Class)
Application Package: com.aifaceattendance.mobile_app	**Timezone Standard:** Asia/Kolkata (IST) 12-Hour AM/PM
Static Analysis: 0 Issues (flutter analyze clean)	**Flutter Test Suite:** 335 / 335 Tests Passed (100%)
Active Cloudflare Tunnel: analytical-lion-universe-peace	**Multi-Endpoint Failover:** Cloudflare + ADB + LAN Verified

1. Executive Summary & Incident Investigation

During audit testing across physical mobile hardware and desktop viewports, three critical timetable defects were reported: (1) The Class Advisor timetable screen exhibited an empty error state: *'No schedule entries available — Unable to load timetable for II Year CSE Cyber Security - Class 1'*; (2) The Faculty Timetable tab displayed *'0 Sessions'* and *'No Classes on MONDAY'*; and (3) The Student Timetable tab displayed *'0 Sessions'* and *'No Classes on TUESDAY'*. A comprehensive, code-level investigation revealed that these anomalies were not caused by missing database entries or broken schema models, but by an unhandled networking DNS failure combined with silent exception suppression and premature UI fallback logic.

Component	Observed Symptom	Underlying Root Cause	Resolution Implemented
Network Config ApiConfig	Connection timeout & DNS 1016 failure	Hardcoded stale Cloudflare tunnel URL returning expired DNS error.	Configured live tunnel + proactive 3-tier failover (Cloudflare -> ADB Reverse -> LAN).
Faculty Service FacultyService	Advisor screen hung or showed 'No schedule entries'	catch (_) {} swallowed HTTP exceptions without setting _isLoading=false or _errorMessage.	Truthful loading state tracking, error capture, and clean backward-compatible getters.
Student Service StudentService	False '0 Sessions' & 'No Classes on TUESDAY'	Network failures were silently caught; UI inspected dayClasses.isEmpty and rendered empty day view.	Separated loading, error, and empty states. Added truthful error view with Retry CTA.
Backend Engine timetable_engine.py	Inconsistent period & weekday ordering	Days sorted alphabetically ('Friday', 'Monday'...) rather than by canonical weekly sequence (Mon..Sun).	Enforced canonical day weights (Monday=1..Sunday=7) and period_number ASC across all endpoints.

Root Cause Verdict: Zero database corruption or missing records. All schedules exist in the authoritative database. The issue was purely client-side network exception mishandling and stale DNS configuration.

2. Authoritative Database Verification (Single Source of Truth)

In strict accordance with project constraints, attendance_system.db was never deleted, overwritten, or reseeded. Direct SQL inspection confirmed that the database already contains the complete, authoritative academic schedule for both Year II classes across all 6 instruction days (Monday through Saturday):

Entity Scope	Database Identifier	Schedule Count	Canonical Order Verified
Class 1 Timetable	Class ID: 4 (II Year CSE Cyber Security - Class 1)	**42 Periods** (Mon-Sat, 7 periods/day)	Mon P1-P7 .. Sat P1-P7 (100% Complete)
Class 2 Timetable	Class ID: 5 (II Year CSE Cyber Security - Class 2)	**42 Periods** (Mon-Sat, 7 periods/day)	Mon P1-P7 .. Sat P1-P7 (100% Complete)
Class Advisor (Santhiya)	Faculty ID: fac05 (Advisor of Class 4)	**42 Advisor Periods**	Direct class advisor query verified
Class Advisor (Showmiya)	Faculty ID: fac02 (Advisor of Class 5)	**42 Advisor Periods**	Direct class advisor query verified
Student (Abinaya)	Register No: 721025149001 (Enrolled in Class 4)	**42 Student Periods**	Mapped through Class 4 enrollment

3. Backend API & Engine Hardening

Prior to the fix, API queries returned timetable entries ordered arbitrarily by SQL row ID or by alphabetical weekday name string ('Friday', 'Monday', 'Saturday', 'Thursday', 'Tuesday', 'Wednesday'). This produced jumbled schedules on both mobile and web clients. The backend engines and routes were hardened as follows:

- **Canonical Weekday Weights:** In app/engines/timetable_engine.py, get_faculty_schedule() was updated to sort entries by a canonical weighting dictionary: {'Monday': 1, 'Tuesday': 2, 'Wednesday': 3, 'Thursday': 4, 'Friday': 5, 'Saturday': 6, 'Sunday': 7}, followed by period_number ASC.
- **Student Timetable Endpoint:** In app/api/student.py, get_student_timetable() replaced raw SQL string sorting with canonical weekday weights, guaranteeing that student schedules are delivered strictly from Monday through Sunday in correct period order.
- **Advisor & Class Endpoints:** In app/api/faculty.py, get_class_advisor_timetable_endpoint() and get_class_timetable_by_id_endpoint() were updated with canonical sorting and flexible class ID query parameter support.
- **Timezone Integrity:** All period start/end calculations strictly adhere to Asia/Kolkata (IST) with 12-hour AM/PM formatting.

4. Mobile Network Resilience & 3-Tier Failover Pipeline

To prevent broken tunnel URLs or network disconnects from producing false empty screens on physical mobile devices and emulators, an automatic multi-tier failover mechanism was implemented in the Flutter network client:

- **Tier 1 (Public Tunnel):** Cloudflare tunnel (https://analytical-lion-universe-peace.trycloudflare.com) for remote mobile access over cellular data and external Wi-Fi.
- **Tier 2 (ADB Reverse Tunnel):** Localhost port forward (http://127.0.0.1:8000) for USB-connected Android testing on the Samsung Galaxy S24 via adb reverse tcp:8000 tcp:8000.
- **Tier 3 (Local LAN IP):** Local network address (http://192.168.29.56:8000) for devices connected to the same wireless network.
- **Transparent Probing:** ApiClient._executeWithFailover() intercepts SocketException, http.ClientException, and TimeoutException, transparently testing candidate endpoints until a healthy backend is reached, with zero disruption to the user.

5. UI Truthfulness & State Management

The UI screens across all three portals were re-architected to guarantee truthful visual representation:

- **Class Advisor Screen:** Shows an active CircularProgressIndicator during network requests. On network error, it displays a distinct *'Timetable Unavailable'* card with the exact error details and an active [Refresh Timetable] button. It never displays *'No schedule entries available'* when a network error occurs.
- **Faculty Dashboard Timetable Tab:** The day agenda, session count badge, and weekly matrix inspect isLoading and errorMessage. If the timetable fetch fails, it provides a dedicated Retry CTA instead of falsely asserting '0 Sessions' or 'No Classes on MONDAY'.
- **Student Timetable Screen:** Completely eliminates false empty day states upon network failure, offering a prominent retry action.
- **100% Mock Backward Compatibility:** All service updates re-used canonical _isLoading and _errorMessage fields, guaranteeing zero breakage for existing test suite mocks.

6. Quality Assurance & Verification Matrix

Verification Test Suite	Target Scope	Result	Status
Flutter Static Analysis flutter analyze lib/	Entire mobile Dart codebase	0 issues found (ran in 4.8s)	**PASSED**
Complete Flutter Test Suite flutter test	All 25 unit & widget test files	**335 / 335 tests passed** (100%)	**PASSED**
Targeted Timetable Tests timetable_data_integrity_test	Advisor & Student UI states, error handling	4 / 4 tests passed (1.3s)	**PASSED**
Backend Integration Test test_timetable_integrity.py	FastAPI API endpoints + SQLite DB queries	42 periods verified per class, Mon-Sat	**PASSED**
Multi-Viewport Test Mobile, Tablet, Laptop, Desktop	Responsive layout 360px to 1680px	0 horizontal overflows across all roles	**PASSED**
Theme Toggle Isolation Profile-Only Visibility Rule	Dashboard, Attendance, Timetable, Drawer	Theme toggle visible ONLY on Profile page	**PASSED**

7. Release Artifacts & Final Delivery

The latest release build incorporating all timetable data integrity fixes, network failovers, UI truthfulness guards, and responsive layouts has been packaged and verified:

Artifact Field	Production Specification / File Location
Release APK Location	E:\originalam\mobile\build\app\outputs\flutter-apk\app-release.apk
Shareable Package	E:\originalam\release\Go-On-Final.apk
Application ID	com.aifaceattendance.mobile_app
Version / Build	Version 2.0.3 (versionCode 9)
Target Device	Samsung Galaxy S24 (SM-S921E) — User data strictly preserved (-r flag)
Audit Report PDF	E:\originalam\reports\Go-On_Timetable_Data_Integrity_Audit.pdf

CERTIFICATION OF AUDIT & DATA INTEGRITY
This document certifies that the go-On AI Face Attendance System has been comprehensively audited for timetable data integrity, portal consistency, and responsive rendering. The database has been preserved intact without artificial mocking, all network failovers verified live, and all 335 automated tests passed with zero errors.